```
!pip install nltk spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/pyt
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/pyt
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/pytho
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.1
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/loca
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.1
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/pytho
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/d
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/pyth
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_cor
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 97.3 MB/s eta 0:00:00
✔ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import nltk
import spacy
```

```
from nltk.tokenize import sent_tokenize, word_tokenize
from nltk.stem import PorterStemmer
```

```
nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
True
```

medical_text = """ Hypertension is a chronic medical condition characterized by elevated blood pressure. Patients with persistent hypertension are at increased risk of cardiovascular disease, stroke, and kidney failure. Lifestyle modifications such as reduced salt intake, regular physical activity, and medication adherence are essential for effective management. Early diagnosis and continuous monitoring significantly improve patient outcomes. ""

SENTENCE TOKENIZATION USING NLTK sentences = sent_tokenize(medical_text) print("Sentence Tokenization:") for s in sentences: print("-", s)

Sentence Tokenization:- Hypertension is a chronic medical condition characterized by elevated blood pressure.- Patients with persistent hypertension are at increased risk of cardiovascular disease, stroke, and kidney failure.- Lifestyle modifications such as reduced salt intake, regular physical activity, and medication adherence are essential for effective management.- Early diagnosis and continuous monitoring significantly improve patient outcom

WORD TOKENIZATION USING NLTL

```
import nltk
from nltk.tokenize import word_tokenize

medical_text = """
Hypertension is a chronic medical condition characterized by elevated blood pre
Patients with persistent hypertension are at increased risk of cardiovascular c
stroke, and kidney failure. Lifestyle modifications such as reduced salt intake
regular physical activity, and medication adherence are essential for effective
Early diagnosis and continuous monitoring significantly improve patient outcome
"""

words = word_tokenize(medical_text)
print("\nWord Tokenization:")
print(words)
```

```
Word Tokenization:
['Hypertension', 'is', 'a', 'chronic', 'medical', 'condition', 'characterized',
```

LEMMATIZATION USING spaCY

```
nlp = spacy.load("en_core_web_sm")
doc = nlp(medical_text)
lemmatized_words = [token.lemma_ for token in doc if token.is_alpha]
print("Lemmatized Words:")
print(lemmatized_words)
```

```
Lemmatized Words:
['Hypertension', 'be', 'a', 'chronic', 'medical', 'condition', 'characterize', '
```

COMPARISION : ORIGINALvs STEMMED vs LEMMATIZED

```
import nltk
from nltk.stem import PorterStemmer

ps = PorterStemmer()
comparison = []
for word in words:
    if word.isalpha():
        stem = ps.stem(word)
        lemma = nlp(word)[0].lemma_
        comparison.append((word, stem, lemma))
comparison[:15]
```

```
[('Hypertension', 'hypertens', 'hypertension'),
 ('is', 'is', 'be'),
 ('a', 'a', 'a'),
 ('chronic', 'chronic', 'chronic'),
 ('medical', 'medic', 'medical'),
 ('condition', 'condit', 'condition'),
 ('characterized', 'character', 'characterize'),
 ('by', 'by', 'by'),
 ('elevated', 'elev', 'elevate'),
 ('blood', 'blood', 'blood'),
 ('pressure', 'pressur', 'pressure'),
 ('Patients', 'patient', 'patient'),
 ('with', 'with', 'with'),
 ('persistent', 'persist', 'persistent'),
 ('hypertension', 'hypertens', 'hypertension')]
```

SRUniversity="""The SR University campus is located in Ananthasagar village of Hasanparthy Mandal in Warangal, Telan It is in 150 acres, with both separate hostel facilities for boys and girls. There is a huge central library along with Indias largest Technology Business Incubator (TBI) in tier 2 cities."""

SRUniversity ' The SR University campus is located in Ananthasagar village of Hasanparthy Mandal in Warangal, Telangana, India. \nI t is in 150 acres, with both separate hostel facilities for boys and girls. \nThere is a huge central library along w

```
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')
from nltk.tokenize import word_tokenize

SRUniversity="""The SR University campus is located in Ananthasagar village of
It is in 150 acres, with both separate hostel facilities for boys and girls.
There is a huge central library along with Indias largest Technology Business I

word_tokenize(SRUniversity)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
['The',
 'SR',
 'University',
 'campus',
 'is',
 'located',
 'in',
 'Ananthasagar',
 'village',
 'of',
 'Hasanparthy',
 'Mandal',
 'in',
 'Warangal',
 ',',
 'Telan',
 'It',
 'is',
 'in',
 '150',
 'acres',
 ',',
 'with',
 'both',
 'separate',
 'hostel',
 'facilities',
 'for',
 'boys',
 'and',
```

```
 'girls',
 '.',
 'There',
 'is',
 'a',
 'huge',
 'central',
 'library',
 'along',
 'with',
 'Indias',
 'largest',
 'Technology',
 'Business',
 'Incubator',
 '(',
 'TBI',
 ')',
 'in',
 'tier',
 '2',
 'cities',
 '.']
```

```
from nltk.tokenize import sent_tokenize
sent_tokenize(SRUniversity)
```

```
['The SR University campus is located in Ananthasagar village of Hasanparthy Mandal in Warangal, Telan\nIt is in 150 acres, with both separate hostel facilities for boys and girls.',
 'There is a huge central library along with Indias largest Technology Business Incubator (TBI) in tier 2 cities.']
```

```
nltk.download("stopwords")
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

```
words_in_quote = word_tokenize(SRUniversity)
words_in_quote
```

```
['The',
 'SR',
 'University',
 'campus',
 'is',
 'located',
 'in',
 'Ananthasagar',
 'village',
 'of',
```

```
 'Hasanparthy',
 'Mandal',
 'in',
 'Warangal',
 ',',
 'Telan',
 'It',
 'is',
 'in',
 '150',
 'acres',
 ',',
 'with',
 'both',
 'separate',
 'hostel',
 'facilities',
 'for',
 'boys',
 'and',
 'girls',
 '.',
 'There',
 'is',
 'a',
 'huge',
 'central',
 'library',
 'along',
 'with',
 'Indias',
 'largest',
 'Technology',
 'Business',
 'Incubator',
 '(',
 'TBI',
 ')',
 'in',
 'tier',
 '2',
 'cities',
 '.']
```

```
stop_words = set(stopwords.words("english"))
filtered_list = []
for word in words_in_quote:
    if word.casefold() not in stop_words:
        filtered_list.append(word)
print(filtered_list)
```

```
['SR', 'University', 'campus', 'located', 'Ananthasagar', 'village', 'Hasanparth
```

```
import nltk
nltk.download('omw-1.4')
nltk.download('wordnet')
from nltk.stem import WordNetLemmatizer
lemmatizer = WordNetLemmatizer()
words = word_tokenize(SRUniversity)
for word in words:
    print(word,"--->",lemmatizer.lemmatize(word))
```

```
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Downloading package wordnet to /root/nltk_data...
The ---> The
SR ---> SR
University ---> University
campus ---> campus
is ---> is
located ---> located
in ---> in
Ananthasagar ---> Ananthasagar
village ---> village
of ---> of
Hasanparthy ---> Hasanparthy
Mandal ---> Mandal
in ---> in
Warangal ---> Warangal
, ---> ,
Telan ---> Telan
It ---> It
is ---> is
in ---> in
150 ---> 150
acres ---> acre
, ---> ,
with ---> with
both ---> both
separate ---> separate
hostel ---> hostel
facilities ---> facility
for ---> for
boys ---> boy
and ---> and
girls ---> girl
. ---> .
There ---> There
is ---> is
a ---> a
huge ---> huge
central ---> central
library ---> library
along ---> along
with ---> with
Indias ---> Indias
largest ---> largest
Technology ---> Technology
Business ---> Business
```

```
Incubator ---> Incubator
( ---> (
TBI ---> TBI
) ---> )
in ---> in
tier ---> tier
2 ---> 2
cities ---> city
. ---> .
```

```
from nltk.stem import RegexpStemmer
regexp = RegexpStemmer('ing|e', min=4)
words = word_tokenize(SRUniversity)
for word in words:
    print(word,"--->",regexp.stem(word))
```

```
The ---> The
SR ---> SR
University ---> Univrsity
campus ---> campus
is ---> is
located ---> locatd
in ---> in
Ananthasagar ---> Ananthasagar
village ---> villag
of ---> of
Hasanparthy ---> Hasanparthy
Mandal ---> Mandal
in ---> in
Warangal ---> Warangal
, ---> ,
Telan ---> Tlan
It ---> It
is ---> is
in ---> in
150 ---> 150
acres ---> acrs
, ---> ,
with ---> with
both ---> both
separate ---> sparat
hostel ---> hostl
facilities ---> facilitis
for ---> for
boys ---> boys
and ---> and
girls ---> girls
. ---> .
There ---> Thr
is ---> is
a ---> a
huge ---> hug
central ---> cntral
library ---> library
along ---> along
```

```
with ---> with
Indias ---> Indias
largest ---> largst
Technology ---> Tchnology
Business ---> Businss
Incubator ---> Incubator
( ---> (
TBI ---> TBI
) ---> )
in ---> in
tier ---> tir
2 ---> 2
cities ---> citis
. ---> .
```

```
from nltk import LancasterStemmer
Lanc = LancasterStemmer()
words = word_tokenize(SRUniversity)
for word in words:
    print(word,"--->",Lanc.stem(word))
```

```
The ---> the
SR ---> sr
University ---> univers
campus ---> camp
is ---> is
located ---> loc
in ---> in
Ananthasagar ---> ananthasag
village ---> vil
of ---> of
Hasanparthy ---> hasanparthy
Mandal ---> mand
in ---> in
Warangal ---> warang
, ---> ,
Telan ---> tel
It ---> it
is ---> is
in ---> in
150 ---> 150
acres ---> acr
, ---> ,
with ---> with
both ---> both
separate ---> sep
hostel ---> hostel
facilities ---> facil
for ---> for
boys ---> boy
and ---> and
girls ---> girl
. ---> .
There ---> ther
is ---> is
```

```
a ---> a
huge ---> hug
central ---> cent
library ---> libr
along ---> along
with ---> with
Indias ---> india
largest ---> largest
Technology ---> technolog
Business ---> busy
Incubator ---> incub
( ---> (
TBI ---> tbi
) ---> )
in ---> in
tier ---> tier
2 ---> 2
cities ---> city
. ---> .
```

PDF ASSIGNMENT:

"NLP models are transforming the world rapidly!"

1)IN WORD TOKENS :

using nltk:

```
import nltk
from nltk.tokenize import word_tokenize

# Download tokenizer (run once)
nltk.download('punkt')
nltk.download('punkt_tab') # Added to download the required resource

text = "NLP models are transforming the world rapidly!"

tokens = word_tokenize(text)
print(tokens)
```

```
['NLP', 'models', 'are', 'transforming', 'the', 'world', 'rapidly', '!']
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

```
['NLP', 'models', 'are', 'transforming', 'the', 'world', 'rapidly', '!']
```

```
['NLP', 'models', 'are', 'transforming', 'the', 'world', 'rapidly', '!']
```

using spacy

```
import spacy

# Load English model
nlp = spacy.load("en_core_web_sm")

text = "NLP models are transforming the world rapidly!"

doc = nlp(text)
tokens = [token.text for token in doc if not token.is_punct]

print(tokens)
```

```
['NLP', 'models', 'are', 'transforming', 'the', 'world', 'rapidly']
```

2)STEMMED WORDS:

USING NLTK:

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer

# Download tokenizer (run once)
nltk.download('punkt')

text = "NLP models are transforming the world rapidly!"

# Tokenization
tokens = word_tokenize(text)

# Initialize stemmer
stemmer = PorterStemmer()

# Apply stemming (ignore punctuation)
stemmed_words = [stemmer.stem(word) for word in tokens if word.isalpha()]

print(stemmed_words)
```

```
['nlp', 'model', 'are', 'transform', 'the', 'world', 'rapidli']
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
```

3)Lemmatization :

```
import spacy

# Load spaCy English model
nlp = spacy.load("en_core_web_sm")

text = "NLP models are transforming the world rapidly!"

# Process text
doc = nlp(text)

# Apply lemmatization (ignore punctuation)
lemmatized_words = [token.lemma_ for token in doc if token.is_alpha]

print(lemmatized_words)
```

```
['NLP', 'model', 'be', 'transform', 'the', 'world', 'rapidly']
```